Olist e-commerce data warehouse

From 9 raw CSV files to a star schema, 126 automated quality checks and five decision-ready findings

Project ds4fm2t3 | DuckDB, Python, Dagster, Streamlit

Growth is strong, but retention and delivery speed cap it

BRL 13.5M
revenue from 98,199 valid orders

+138%
Jan-Aug 2018 revenue vs Jan-Aug 2017

3.04%
of customers ever buy a second time

2.3 vs 4.3
review score: late vs on-time delivery

Pipeline: 126 data-quality checks, 0 failures; 9 warnings trace to known source-data anomalies.

Recommendation: fund a retention programme and a regional logistics fix before more acquisition spend.

One pipeline, three layers, one schedule



Dagster schedule `0 4 * * *` - ingest, clean, incremental ELT, data-quality gate

Incremental by design: row-hash change detection inserts new rows and updates changed ones; a re-run with no new data changes nothing.

Proven by test: an initial partial load plus an incremental top-up equals a one-shot full load.

Four facts share six conformed dimensions

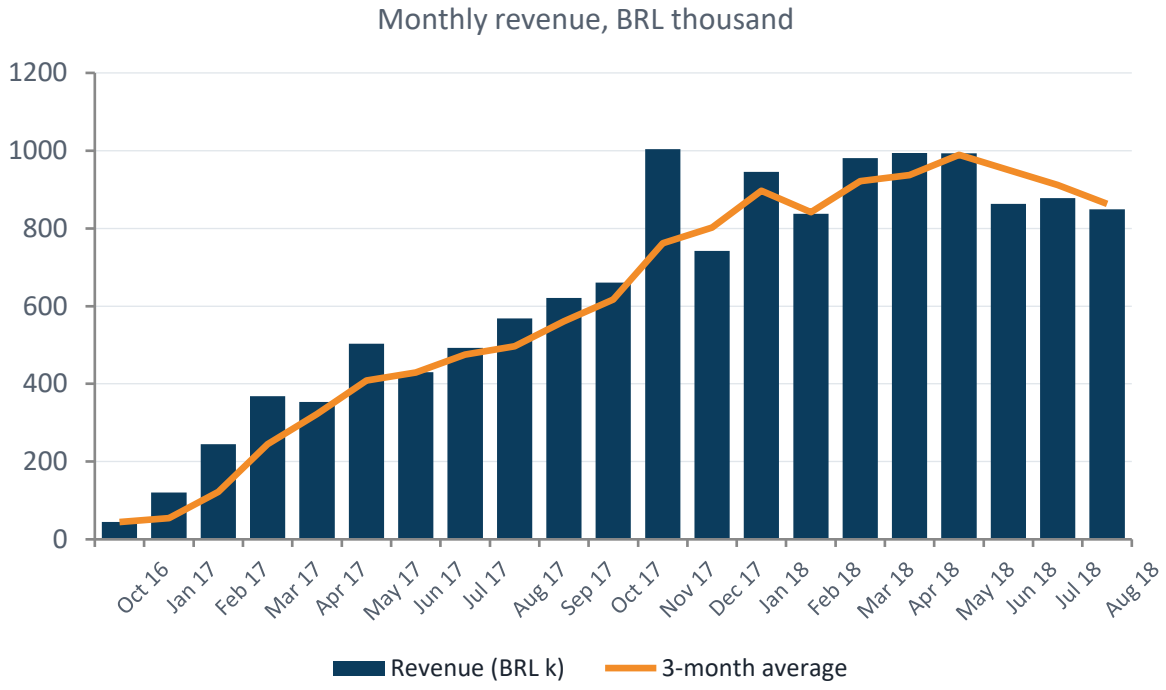
F_ORDER_ITEMS	F_ORDERS	F_ORDER_PAYMENTS	F_ORDER_REVIEWS
1 row per order item	1 row per order	1 row per payment	1 row per review

D_DATE	D_CUSTOMER	D_PRODUCT	D_SELLER	D_GEOGRAPHY	D_PAYMENT_TYPE

Why this shape: each fact has one grain and additive measures; customers are keyed on the real person (unique id), not the per-order id, so lifetime value is correct.

Derived columns: total_sale_amount, customer_lifetime_value, delivery_days, is_late, review flags and more.

Revenue grew 138% year on year, then plateaued



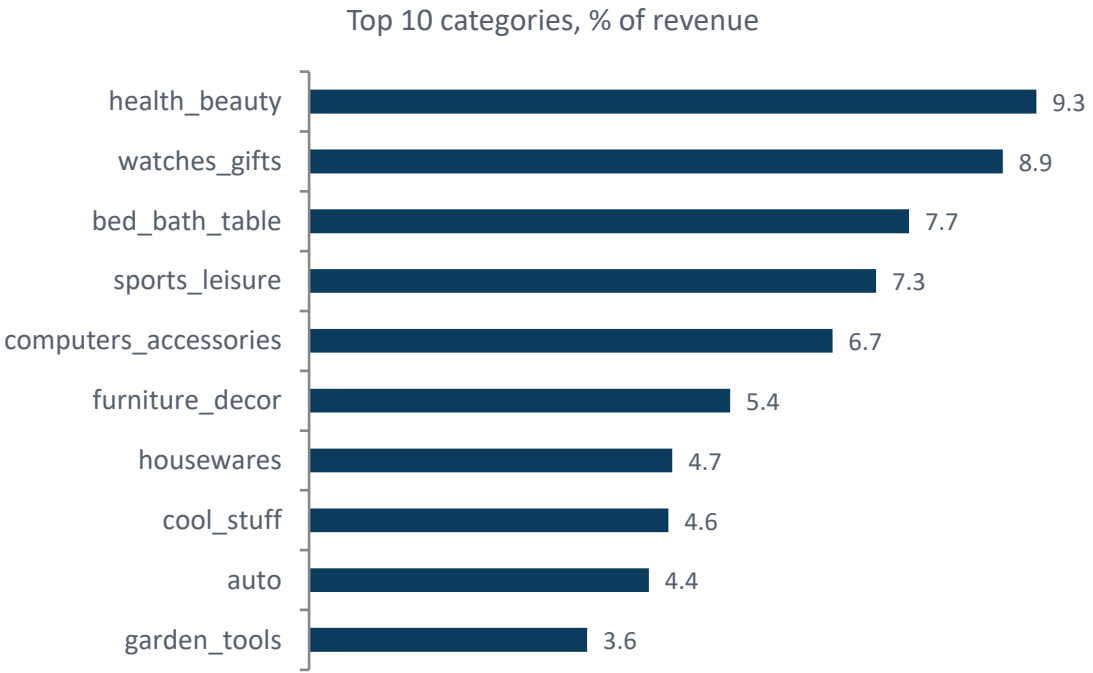
Peak: 2017-11
7,421 orders, BRL 1,003,862

Since early 2018
Monthly revenue has stayed between BRL 0.84M and 0.99M; growth has stalled.

Data note
Sep-2016, Dec-2016 and Sep-2018 hold under 100 orders and are excluded.

Code: findings/f1_monthly_sales_trend.py

Demand is broad across categories, concentrated in sellers



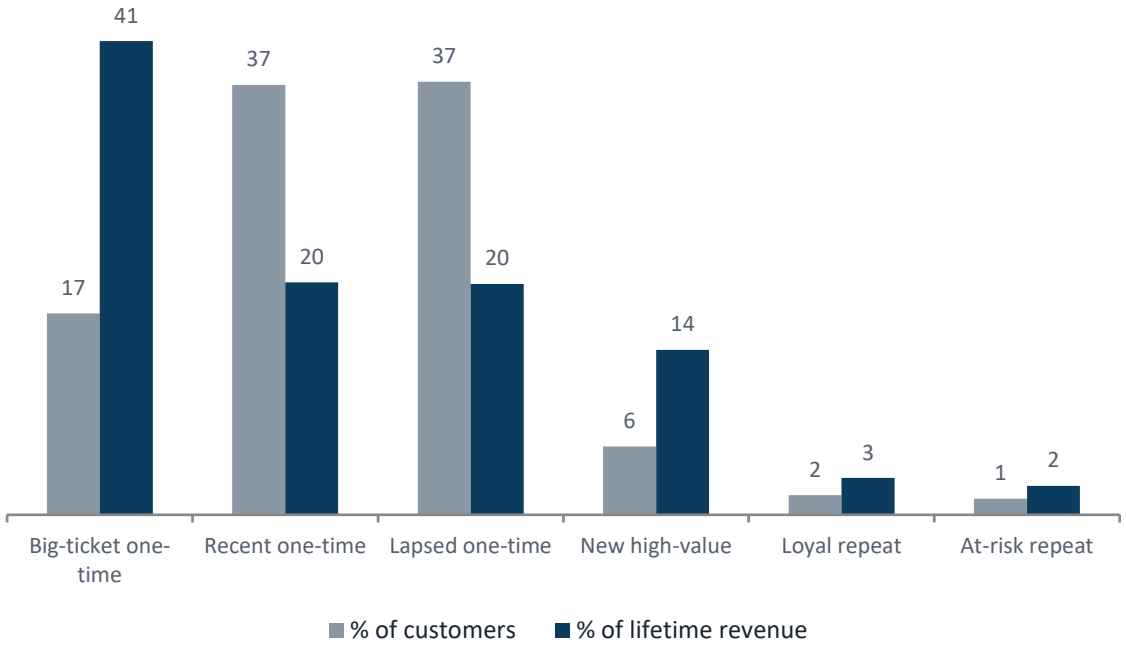
67%

of revenue comes from the top 10% of 3,053 sellers

No category exceeds 10% of revenue; the top five together make up 40%.

Code: findings/f2_top_products.py

97% of customers buy once; one-time big spenders matter most



Big-ticket one-time buyers
17% of customers deliver 41% of revenue.

Repeat buyers
Worth BRL 308 each vs BRL 161 for one-time buyers, yet only 3.04% of customers.

Code: findings/f3_customer_segmentation.py

Late delivery halves the review score



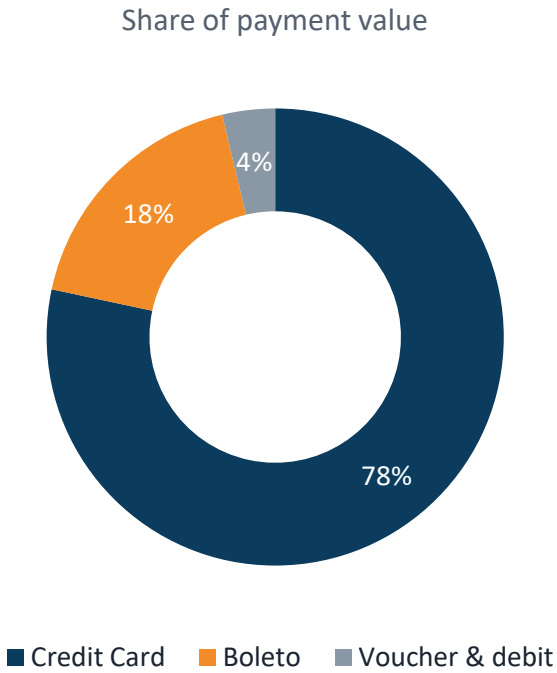
4.3 → 2.3

average score, on-time vs late; 62% of late orders get a 1-2 star review (9% when on time)

Distance matters: AL waits 25 days on average, SP just 9.

Code: findings/f4_delivery_and_reviews.py

Cards dominate payment; the Southeast dominates demand



78%

of value paid by credit card, in 3.5 instalments on average

65%

of revenue from the Southeast; SP alone is 38%

Instalments are widely used, so card-partner offers are worth testing as a conversion lever.

Code: findings/f5_payments_and_geography.py

126 automated checks pass; 9 warnings are source-data defects

Source anomaly found by the tests	Rows
Carrier hand-over dated before approval	1,358
Payments differ from items + freight by more than 1.00	250
Carrier hand-over dated before purchase	161
Delivery dated before carrier hand-over	23
Order marked delivered but no delivery date	8

How it is handled

Hard rules (keys, grain, totals, referential integrity) fail the run. Known source defects are reported, not hidden, and never silently dropped.

Results are stored in FDL_DATA.DQ_RESULT_LOG and shown in the dashboard.

Code: pipeline/dq_tests.py

What to do next

1. **Launch a retention programme**

   Move the 3.04% repeat rate: target big-ticket one-time buyers with a second-purchase offer.

2. **Fix regional delivery promises**

   Re-set estimates and add carriers for the North and Northeast, where waits are 2-3x longer.

3. **Reduce seller concentration risk**

   Onboard and develop mid-tier sellers; two thirds of revenue sits with the top 10%.